

EvidenceMem: A Confirmatory Study of Reliability-Weighted Prototype Memory for Frozen Vision–Language Encoders

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Abstract

External memories let a frozen vision–language encoder adapt to a labeled image collection without updating model weights, but a full nearest-neighbor index grows with the collection. We study whether a small memory of real-image prototypes can preserve classification accuracy and whether prototype reliability improves an equal-budget coverage baseline. EvidenceMem overclusters each class, selects 40 medoids per class by facility coverage, scores each medoid using compactness, neighborhood purity, and text alignment, and combines class-conditional retrieval with text scores. The encoder, split, method, memory budget, seeds, and two hypotheses were frozen after development. On an untouched 3,300-image confirmatory split drawn from 11 categories of 512-pixel source images, SigLIP 2 features and a 440-image memory reach $94.93\% \pm 0.22\%$ accuracy. Full 10-nearest-neighbor classification reaches 95.91% with 13,200 stored images. The point gap is -0.982 percentage points, but the hierarchical paired 95% interval, $[-1.285, -0.679]$, crosses the predeclared -1 -point boundary; non-inferiority is therefore not established. EvidenceMem exceeds an equal-count facility baseline by 0.121 points, with an interval of $[-0.103, 0.352]$, so the reliability hypothesis is also unsupported. We release the executed notebook, raw predictions, split manifest, and independent metric audit. The result supports bounded prototype memory as a useful accuracy–storage tradeoff on this dataset, while showing that the tested reliability mechanism does not yet justify a stronger claim.

1 Introduction

Contrastive vision–language encoders turn images and class descriptions into a shared representation space. Their zero-shot behavior is useful, but labeled examples often remain the most direct way to adapt a system to a particular image collection. A full nearest-neighbor index can exploit every labeled embedding, yet its storage and search cost grow linearly with the support set. A prototype memory offers a simple alternative: retain a small set of real examples and classify a query by comparing it with those examples. Because each prototype is an actual source image, the system can also return the retrieved items for inspection.

Compression creates two different problems. The selected prototypes must cover the visual variation within each class, and the scoring rule must avoid over-trusting

weak or atypical prototypes. Our earlier implementation mixed these concerns inside the selection objective. Development experiments suggested that reliability-aware selection could sacrifice useful coverage. The present version therefore freezes the support set produced by a coverage-only facility objective and uses reliability only when scoring the selected prototypes and combining visual and textual predictions. This distinction matters: the experiment tests reliability-aware scoring on a matched memory, not a new reliability-aware coreset guarantee.

We evaluate this design under a confirmatory protocol rather than reporting the best configuration found after examining the test set. Development data selected the SigLIP 2 ViT-B/16 encoder at 384-pixel input, the continuous-fusion variant, five memory seeds, and a default budget of 40 images per class. The confirmatory notebook hard-codes that choice and evaluates a previously untouched split. The primary hypothesis asks whether the compressed memory is non-inferior to full kNN at a one-percentage-point margin. A secondary hypothesis asks whether reliability-aware scoring improves an equal-count facility baseline.

We contribute a frozen, audited comparison of a 440-image prototype memory with a 13,200-image kNN index. The memory comes within 0.982 points of kNN on this split. However, the result fails the predeclared non-inferiority test and shows no clear gain over the matched facility baseline. Figure 1 places the result beside the other frozen baselines. The negative decisions are scientifically useful. They separate a promising compression point estimate from conclusions that the data do not support. The error analysis also shows where the remaining errors are concentrated.

2 Related Work

Frozen vision–language representations. CLIP demonstrated that natural-language supervision can produce image representations that transfer through text-defined classifiers [3]. SigLIP 2 extends this family with a revised training recipe and stronger transfer and localization behavior [4]. We use a frozen SigLIP 2 checkpoint as a feature extractor and do not fine-tune its parameters. The question is therefore about the external memory and scoring rule, not representation learning.

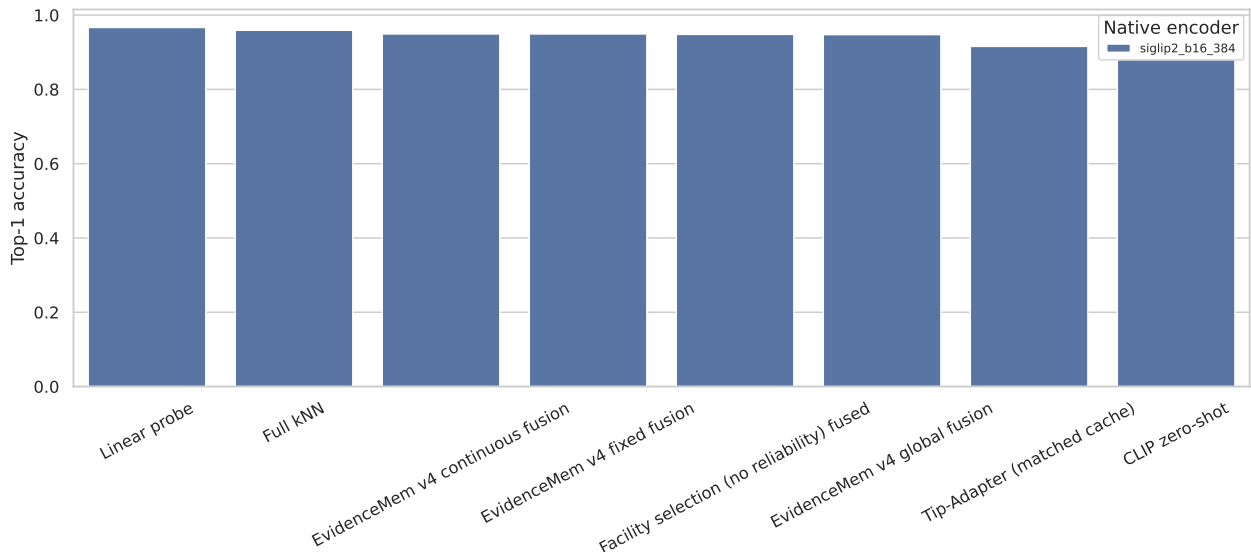


Figure 1: Confirmatory top-1 accuracy on the frozen 3,300-image split. Bars summarize five memory seeds where a method is stochastic. The linear probe and full kNN retain an accuracy advantage; EvidenceMem uses 440 stored images rather than the 13,200-image kNN support set.

Cache-based adaptation. Tip-Adapter augments text logits with similarities to a labeled feature cache, providing a training-free few-shot adaptation rule [5]. ProKeR later interpreted cache adapters as local kernel methods and argued for incorporating global information [1]. EvidenceMem shares the use of a frozen feature space and a labeled cache, but imposes an exact per-class storage budget and retains source medoids. We include Tip-Adapter with the same 440-example cache budget. This experiment does not claim a new state of the art in few-shot adaptation because it covers one curated dataset and one selected encoder.

Coverage selection. Facility-location objectives reward a set whose members are similar to the remaining data. With nonnegative similarities, the usual maximum-coverage form is monotone and submodular, and cardinality-constrained greedy maximization has the standard approximation result [2]. That property concerns coverage of the frozen embedding set; it is not an accuracy guarantee. In our confirmatory method, facility coverage determines the stored support set and reliability changes only its downstream scores.

3 Method

3.1 Task and frozen representations

Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$ be a labeled support set with C classes. A frozen image encoder f produces a unit-normalized vector $z_i = f(x_i)/\|f(x_i)\|_2$. A frozen text encoder produces one class vector t_c by averaging and normalizing six prompted descriptions of class c . The memory budget is B real images per class, so the complete memory

contains BC prototype vectors and their source identifiers.

3.2 Coverage-selected real-image prototypes

For each class, we first fit $2B$ MiniBatchKMeans clusters to its support vectors. The candidate for a cluster is the member nearest its normalized centroid, which guarantees that every candidate maps to a displayable source image. We then greedily select B candidates. If V_c denotes all support vectors in class c and S_c a candidate set, the selection objective is

$$F_c(S_c) = \frac{1}{|V_c|} \sum_{z \in V_c} \max_{p \in S_c} \frac{1 + z^\top p}{2}. \quad (1)$$

The transform maps cosine similarity to $[0, 1]$. Greedy selection starts from an empty set and adds the candidate with the largest marginal gain until $|S_c| = B$. The same selected indices are used by the facility control and the v4 treatment.

3.3 Prototype reliability

Each selected prototype p_j receives three validation-tunable signals. Compactness q_j is its mean cosine similarity to the members of its original cluster. Purity u_j is the fraction of its nearest support neighbors that share its label. Text alignment $a_j = p_j^\top t_{y_j}$ measures agreement with the prompted class vector. Compactness and text alignment are mapped from $[-1, 1]$ to $[0, 1]$; purity already lies on that interval. For nonnegative component weights

w_q, w_u, w_a , reliability is

$$r_j = \text{clip}\left(\frac{w_q \tilde{q}_j + w_u u_j + w_a \tilde{a}_j}{w_q + w_u + w_a}, 0.05, 1\right). \quad (2)$$

If an alignment is unavailable, its term is excluded from both numerator and denominator. The component weights are selected on the validation split for each memory seed.

3.4 Class-conditional scoring and fusion

Global retrieval can exclude a class entirely when its prototypes do not enter the overall top- k . We instead retrieve the k_c most similar prototypes inside every class. For query vector z , class c receives the log-mean-exp score

$$\ell_c^{\text{vis}}(z) = \log \frac{1}{k_c} \sum_{j \in \mathcal{N}_{k_c}^c(z)} \exp\left(\frac{z^\top p_j}{\tau} + \gamma \log r_j\right), \quad (3)$$

where $\tau = 0.07$ and $\gamma \geq 0$ controls reliability weighting. A softmax across classes gives $P_{\text{vis}}(y | z)$. Text probabilities $P_{\text{text}}(y | z)$ come from cosine similarities to the class text vectors.

The selected prototypes also define a query-level reliability $R(z)$: for each class, we average the reliabilities of its retrieved prototypes, then average those class values using P_{vis} . After standardizing R with validation statistics, the text weight is

$$\alpha(z) = \text{clip}(b + s \hat{R}(z), 0, 1), \quad P = (1 - \alpha)P_{\text{vis}} + \alpha P_{\text{text}}. \quad (4)$$

The validation search chooses $w_q, w_u, w_a, k_c, \gamma, b$, and s . A final temperature is also selected on validation negative log likelihood. Four of five seeds selected $s = 0$; the adaptive gate therefore collapsed to a constant in most runs. We report this because it limits the evidence for query-adaptive fusion.

4 Confirmatory Protocol

4.1 Data and splits

The source is the Kaggle collection *130k Images 512x512 Universal Image Embeddings*.¹ We use the image files, not precomputed third-party embeddings. The 11 categories are apparel, artwork, cars, dishes, furniture, illustrations, landmark, meme, packaged products, storefronts, and toys. Sampling is balanced: each class contributes 1,200 training, 200 validation, 300 development, and 300 confirmatory images, for 22,000 retained images. Before the split, SHA-256 and perceptual hashes remove four exact, four within-label perceptual, and two cross-label perceptual duplicates from the sampled candidate pool. The split manifest fixes every source path and content hash.

All source files are 512 by 512 pixels. The selected checkpoint, `google/siglip2-base-patch16-384`,

¹<https://www.kaggle.com/datasets/rhtsingh/130k-images-512x512-universal-image-embeddings>

processes them at 384 by 384 pixels using its native preprocessing. We use “512-pixel source images” and “384-pixel encoder input” throughout; treating this experiment as a native 512-pixel encoder comparison would be incorrect.

4.2 Freeze and baselines

Development results fixed protocol revision 2.0.1, sample seed 2026, memory seeds $\{7, 17, 29, 43, 61\}$, the SigLIP 2 checkpoint, the continuous-fusion variant, and $B = 40$. A dedicated confirmatory notebook permits no environment override for the encoder or evaluation split and checks the frozen repository tag and package tree before evaluation. No setting was changed after the confirmatory results were read.

The context baselines are full similarity-weighted kNN, an ℓ_2 -regularized linear probe, and zero-shot text classification. The bounded-memory baselines are random memory, K-means medoids, and equal-count facility selection. We also include Tip-Adapter with an equal-count random cache. Full kNN selects k from $\{3, 5, 10, 20\}$ on validation and chooses $k = 10$. The linear probe selects C from $\{0.01, 0.1, 1, 10\}$ and chooses 10. All methods use the same frozen embeddings and data partitions.

4.3 Hypotheses and uncertainty

The primary hypothesis is that EvidenceMem at $B = 40$ is non-inferior to full kNN with a margin of one accuracy point. Its decision rule requires the lower end of a hierarchical paired 95% interval to exceed -1 point. The bootstrap resamples seeds and then examples within a seed for 5,000 draws. The secondary hypothesis requires the lower end of the same type of interval for EvidenceMem minus facility selection to exceed zero. We also report a two-sided exact sign-flip result across the five seed-level differences. With five seeds, its smallest possible two-sided value is 0.0625, so it is supporting rather than decisive evidence.

Accuracy and macro F1 summarize classification. We report 15-bin expected calibration error (ECE) after validation-only temperature selection and area under the selective risk-coverage curve (AURC). Reported query time covers the memory or classifier scoring stage only; it excludes image decoding, preprocessing, and encoder inference.

4.4 Artifact audit

The run completed on one Tesla T4 with Python 3.12.13 and PyTorch 2.10.0. Its executed notebook contains 23 cells, including 14 code cells with execution counts 1 through 14 and no error output. The published archive contains 91 unique members and passes a full CRC check. An independent script checks all 23 files named by the original run manifest, then loads 65 raw prediction packets. Each packet contains 3,300 labels, predictions, probabilities, and scores. Recomputed accuracy, balanced accuracy, macro F1, negative log likelihood, Brier score, ECE, and AURC

match the CSV rows to a maximum absolute difference of 2.22×10^{-16} .

5 Results

5.1 Main comparison

Table 1 shows the frozen confirmatory result. The linear probe is strongest at 96.67%, followed by full kNN at 95.91%. EvidenceMem reaches 94.93% while retaining 440 images, or one thirtieth of the kNN support set. It improves substantially over random memory and the matched Tip-Adapter implementation, but those comparisons do not rescue either predeclared hypothesis.

5.2 Predeclared decisions

The full-kNN difference is -0.9818 points, which lies just inside the one-point margin as a point estimate. The lower confidence bound is -1.285 points, however, so the required decision rule fails. This is not proof that the method is worse by more than one point; the interval spans the boundary. It is evidence that this experiment does not establish non-inferiority.

Against equal-count facility selection, the mean difference is $+0.1212$ points and the interval includes zero. Seed differences are $+0.303$, $+0.061$, -0.091 , $+0.364$, and -0.030 points. The reliability treatment therefore has a small and inconsistent effect. Table 2 records both decisions.

5.3 Memory budget

Figure 2 and Table 3 show the predeclared budget curve. Accuracy rises monotonically from 20 to 80 prototypes per class. At 80 per class, EvidenceMem reaches 95.44% with a 15-fold smaller support set than full kNN. Facility selection reaches 95.53% at the same budget, so the reliability treatment does not dominate as memory grows. These budget comparisons are exploratory because $B = 40$ was the frozen hypothesis setting.

5.4 Where compression loses accuracy

The kNN gap is not uniform across categories. Averaged across memory seeds, artwork is 5.00 points below full kNN, illustrations is 2.13 points below, and storefronts is 1.00 point below. Apparel and meme are the only categories where EvidenceMem exceeds full kNN, by 0.20 and 0.67 points. Across the five seed-specific prediction files, the most frequent EvidenceMem error is artwork predicted as illustrations (132 cases), followed by artwork as landmark (89) and artwork as meme (51). These labels mix visual style and semantic content, which makes a small class-balanced memory particularly sensitive to boundary examples.

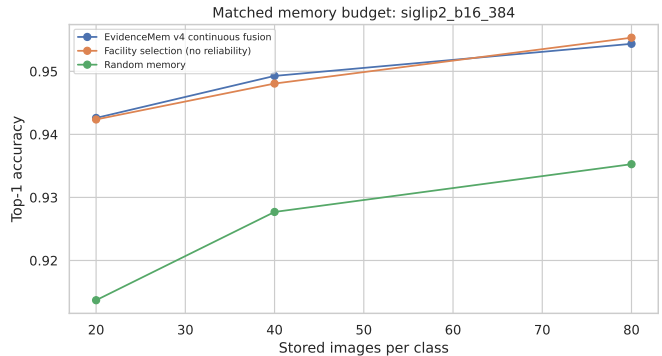


Figure 2: Accuracy versus stored images per class. The 20- and 80-image settings are exploratory; only 40 per class was used for the frozen primary decision.

The mechanism ablation is equally informative. Class-conditional visual scoring alone reaches 94.90%; fixed fusion reaches 94.92%; continuous fusion reaches 94.93%. The three values differ by at most 0.03 points. Four validation runs set the continuous gate slope to zero, and three set its intercept to zero. The present evidence therefore points to class-conditional coverage as the useful part of v4, not a reliable benefit from per-query fusion adaptation.

5.5 Calibration and scoring cost

After validation-only temperature scaling, EvidenceMem has ECE 1.81%, close to the facility baseline’s 1.87% but above full kNN’s 1.20%. Its AURC is 0.00457, compared with 0.00768 for facility selection and 0.01082 for full kNN. This selective-ranking result is secondary and should be confirmed on another dataset before it is treated as a reliability claim.

The measured scoring stage takes 0.060 ms per query for EvidenceMem, 0.018 ms for facility selection, and 0.103 ms for full kNN. Class-conditional retrieval is slower than a single search over the same 440 prototypes, yet still faster than the 13,200-vector kNN search in this implementation. These timings exclude the frozen encoder and were taken inside one Kaggle run, so they describe the experiment rather than deployment latency.

6 Discussion

The result supports a narrow conclusion. Real-image medoids preserve most of full-kNN accuracy while reducing the number of stored support images by a factor of 30. That is a useful operating point for systems that value inspectable examples or bounded storage. It is not enough to call the method non-inferior under the frozen one-point rule. The lower confidence bound misses the boundary by 0.285 points.

The reliability mechanism also needs revision. Validation chooses different reliability components across seeds,

Table 1: Confirmatory classification at the default budget. Values are percentages; $\pm$ is the standard deviation across five memory seeds. Deterministic baselines have zero seed variation. ECE is reported after validation-only temperature scaling.

Method	Stored images	Accuracy	Macro F1	ECE
Linear probe	—	96.67	96.66	1.18
Full kNN	13,200	95.91	95.88	1.20
EvidenceMem continuous fusion	440	94.93 ± 0.22	94.88 ± 0.21	1.81
Facility selection, no reliability	440	94.81 ± 0.10	94.76 ± 0.10	1.87
K-means medoids	440	94.53 ± 0.19	94.46 ± 0.19	2.54
Random memory	440	92.77 ± 0.33	92.67 ± 0.33	3.37
Tip-Adapter, matched cache	440	91.58 ± 0.11	91.38 ± 0.13	4.10
Zero-shot text classifier	0	87.97	87.69	32.49

Table 2: Frozen hypothesis tests. Intervals and deltas are accuracy points.

Comparison	Delta	95% interval	Decision
vs. full kNN	-0.982	$[-1.285, -0.679]$	Not supported
vs. facility	+0.121	$[-0.103, +0.352]$	Not supported

Table 3: Exploratory budget curve. Accuracy is mean $\pm$ standard deviation across five seeds; compression is relative to 13,200 training images.

B/class	Stored	Compression	Accuracy
20	220	$60\times$	94.26 ± 0.14
40	440	$30\times$	94.93 ± 0.22
80	880	$15\times$	95.44 ± 0.17

and the continuous gate usually becomes constant. Reliability scores may still help rank evidence. However, the confirmatory comparison does not show a stable accuracy gain over the same coverage-selected support set. The next experiment should compare class-conditional scoring with and without reliability power. It should keep fusion fixed and register the factorial ablation on a new dataset. The current confirmatory split must remain closed to further tuning.

The linear probe result is another useful boundary. If prediction accuracy is the only goal and storing model parameters is acceptable, a supervised linear head is better on this dataset. Prototype memory serves a different need: bounded collections of source examples that can be retrieved and inspected. A paper about this system should evaluate that operational value directly, perhaps through evidence fidelity or a human error-audit study, rather than treating retrieved neighbors as causal explanations.

7 Limitations

The experiment uses one selected encoder and one balanced 11-class subset. Encoder selection occurred on development data, but broader claims need another encoder family and an external dataset. The source categories are het-

erogeneous and partly stylistic; we did not independently relabel the full 22,000-image manifest. Source images are 512 by 512 pixels, while the encoder consumes 384-pixel inputs, so the study does not measure the effect of native 512-pixel encoding.

The storage comparison counts retained source images, not serialized bytes. A linear probe and an embedding-only index have different storage and privacy properties, which image count alone does not capture. Timing excludes encoder inference and is not a hardware-normalized benchmark. Tip-Adapter was reproduced within the common notebook rather than run from its authors’ repository, although the equation and memory count are matched.

Only five stochastic seeds are available. The hierarchical bootstrap uses example-level pairing, while the exact seed-level test has low resolution. Budget results outside 40 images per class are exploratory. Finally, nearest prototypes are associative evidence. They can expose what the classifier retrieved, but they do not prove why a prediction was made or whether a visual feature is causally responsible.

8 Broader Impact

Bounded memory can reduce the amount of source data retained by an adaptive visual system, but it does not remove privacy, copyright, or consent concerns. The prototypes are deliberately recoverable images; that improves inspection and also makes improper retention visible. Deployments should apply source licensing rules, access controls, deletion procedures, and class-level audits before treating the memory as a durable knowledge store.

Reliability and confidence values can be mistaken for guarantees. Our negative confirmatory result is a reminder that a plausible reliability score may not improve decisions consistently. High-impact applications should not use this prototype as an autonomous decision maker. They need domain-specific validation, calibrated escalation rules, and a way for people to inspect and challenge both the prediction and the stored evidence.

9 Conclusion

We tested a reliability-weighted, class-conditional prototype memory under a frozen confirmatory protocol. The 440-image memory achieved 94.93% accuracy and a 30-fold reduction in stored support images relative to full kNN. Its point estimate was within one percentage point, but the predeclared confidence-bound rule did not establish non-inferiority. Reliability-aware scoring also failed to show a clear advantage over equal-count facility selection. The release preserves both outcomes, along with the executed notebook and raw predictions. The next study should test a simpler, registered scoring ablation on genuinely new data rather than tune this split again.

References

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A Prompt Ensemble

For each class label, the text encoder averages these six templates:

- “a photo whose main category is {class}.”
- “a catalog image of {class}.”
- “a real-world image showing {class} as the main subject.”
- “a wide-view image categorized as {class}.”
- “an online image whose visual domain is {class}.”
- “an image of {class}, not merely a small foreground object.”

Class names are expanded where needed. Examples include “clothing or fashion apparel” and “a packaged retail product.” The exact strings are stored in the notebook source.

B Frozen Hyperparameter Search

The class-specific retrieval grid is $k_c \in \{1, 3, 5, 10\}$. Reliability powers are $\{0, 0.5, 1, 2\}$. Fixed text weights and continuous intercepts are $\{0, 0.25, 0.5, 0.75, 1\}$. Gate slopes are $\{-0.5, -0.25, 0, 0.25, 0.5\}$. Reliability component candidates include each individual component and all pairwise equal mixtures. The grid also includes $(0.35, 0.40, 0.25)$ and $(0.45, 0.35, 0.20)$. Temperature is selected from $\{0.02, 0.03, 0.05, 0.07, 0.10, 0.15, 0.20, 0.50, 1\}$. Selection uses validation accuracy with deterministic tie breaks. The release records the choice for each seed in `selected_hyperparameters.json`.

C Reproducibility Map

The result directory contains the executed notebook, the source-run ZIP, the frozen protocol, split manifest, classification and budget tables, hypothesis decisions, figures, per-class analysis, and a release-level SHA-256 manifest. The verification command is:

```
python scripts/verify_confirmatory_release.py
```

The command checks archive CRCs and hashes, every published file, notebook execution, prediction shapes and probabilities, and all recomputable classification metrics.